

Mechanical Properties of Polymers

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Aims of session & learning outcomes

- Describe the mechanical behaviour of polymers
- Explain how the temperature and time influences their mechanical behavior of polymers
- Describe the viscoelastic behaviour of polymers
- Make graphical representations of stress-strain behaviour of polymer materials
- Describe the various stages in the elastic and plastic deformation of polymers and elastomers

Mechanical properties

<i>Material</i>	<i>Specific Gravity</i>	<i>Tensile Modulus (GPa)</i>	<i>Tensile Strength (MPa)</i>	<i>Yield Strength (MPa)</i>	<i>Elongation at Break (%)</i>
Polyethylene (low density)	0.917–0.932	0.17–0.28	8.3–31.4	9.0–14.5	100–650
Polyethylene (high density)	0.952–0.965	1.06–1.09	22.1–31.0	26.2–33.1	10–1200
Poly(vinyl chloride)	1.30–1.58	2.4–4.1	40.7–51.7	40.7–44.8	40–80
Polytetrafluoroethylene	2.14–2.20	0.40–0.55	20.7–34.5	13.8–15.2	200–400
Polypropylene	0.90–0.91	1.14–1.55	31–41.4	31.0–37.2	100–600
Polystyrene	1.04–1.05	2.28–3.28	35.9–51.7	25.0–69.0	1.2–2.5
Poly(methyl methacrylate)	1.17–1.20	2.24–3.24	48.3–72.4	53.8–73.1	2.0–5.5
Phenol-formaldehyde	1.24–1.32	2.76–4.83	34.5–62.1	—	1.5–2.0
Nylon 6,6	1.13–1.15	1.58–3.80	75.9–94.5	44.8–82.8	15–300
Polyester (PET)	1.29–1.40	2.8–4.1	48.3–72.4	59.3	30–300
Polycarbonate	1.20	2.38	62.8–72.4	62.1	110–150

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Compared to metals:

Density g/cm ³	Tensile modulus GPa	Tensile strength MPa	Yield strength MPa	Elongation %
2.7-8.9	70-400	200-2000	100-1000	1-30

Stress-strain behaviour



- Fracture strengths of polymers $\sim 10\%$ of those for metals

But density also $1/10$, so the specific strengths are similar to metals

- Deformation strains for polymers $> 1000\%$

For most metals, deformation strains $< 10\%$

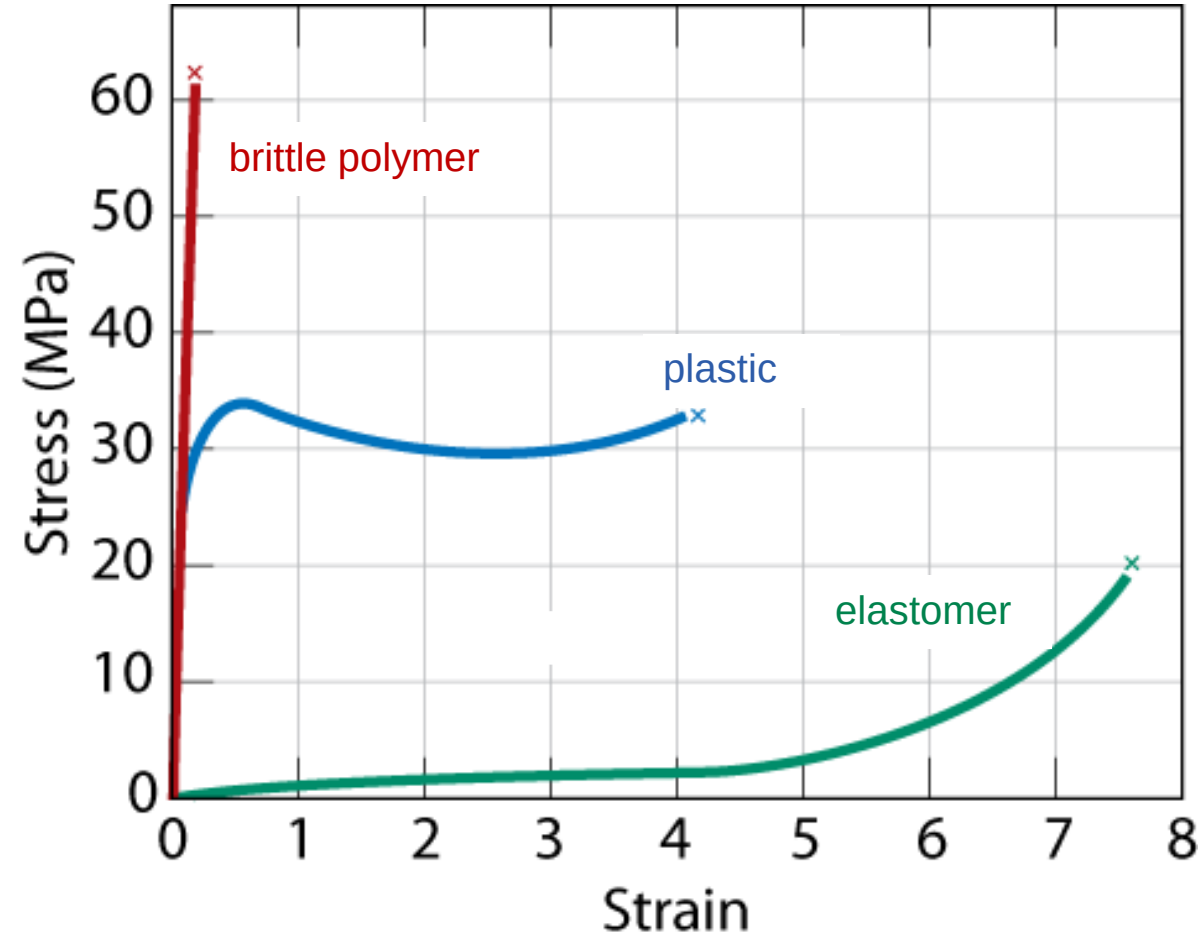


Fig 15.1 Callister & Rethwisch 9e

Mechanisms of deformation

Brittle Crosslinked and Network Polymers

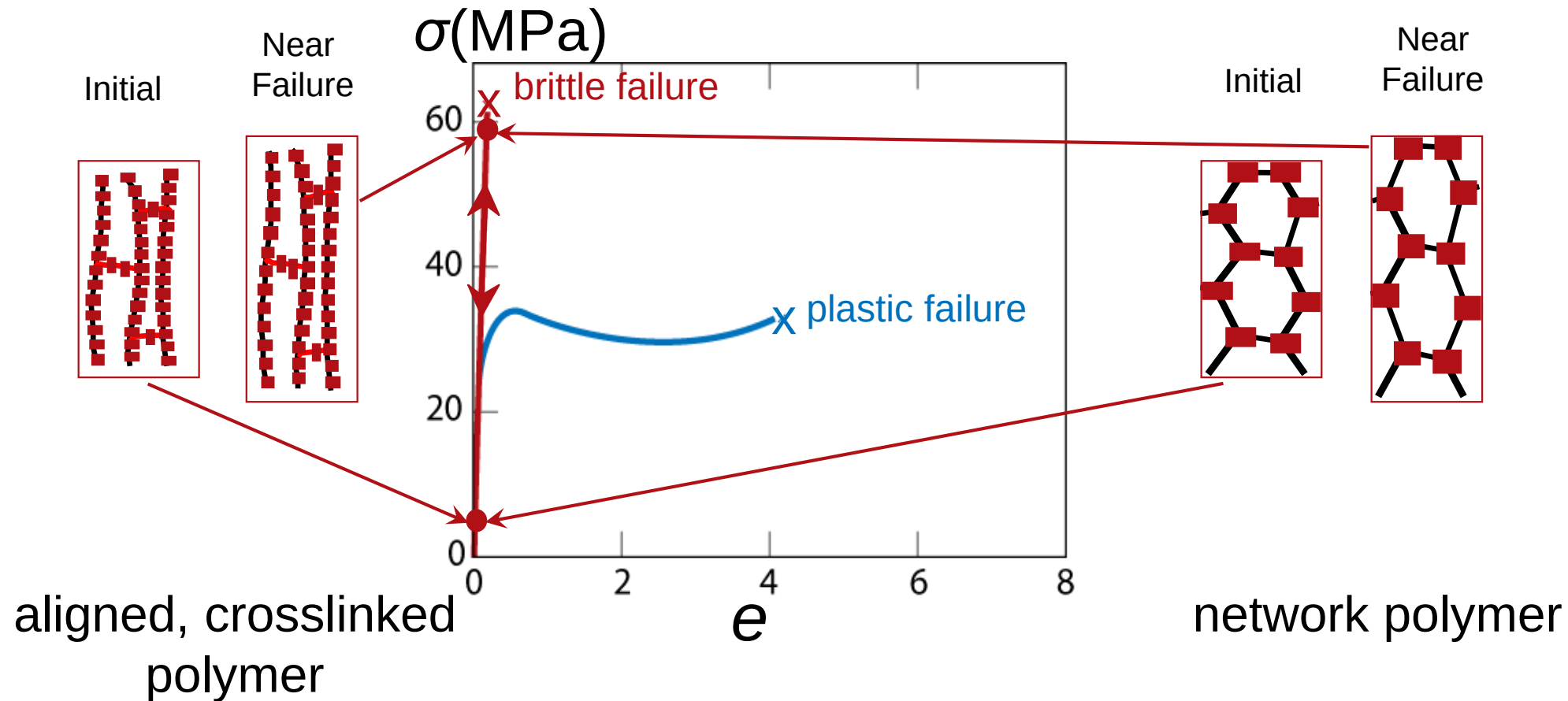


Fig 15.1 Callister & Rethwisch 9e

Mechanisms of deformation

Elastomers

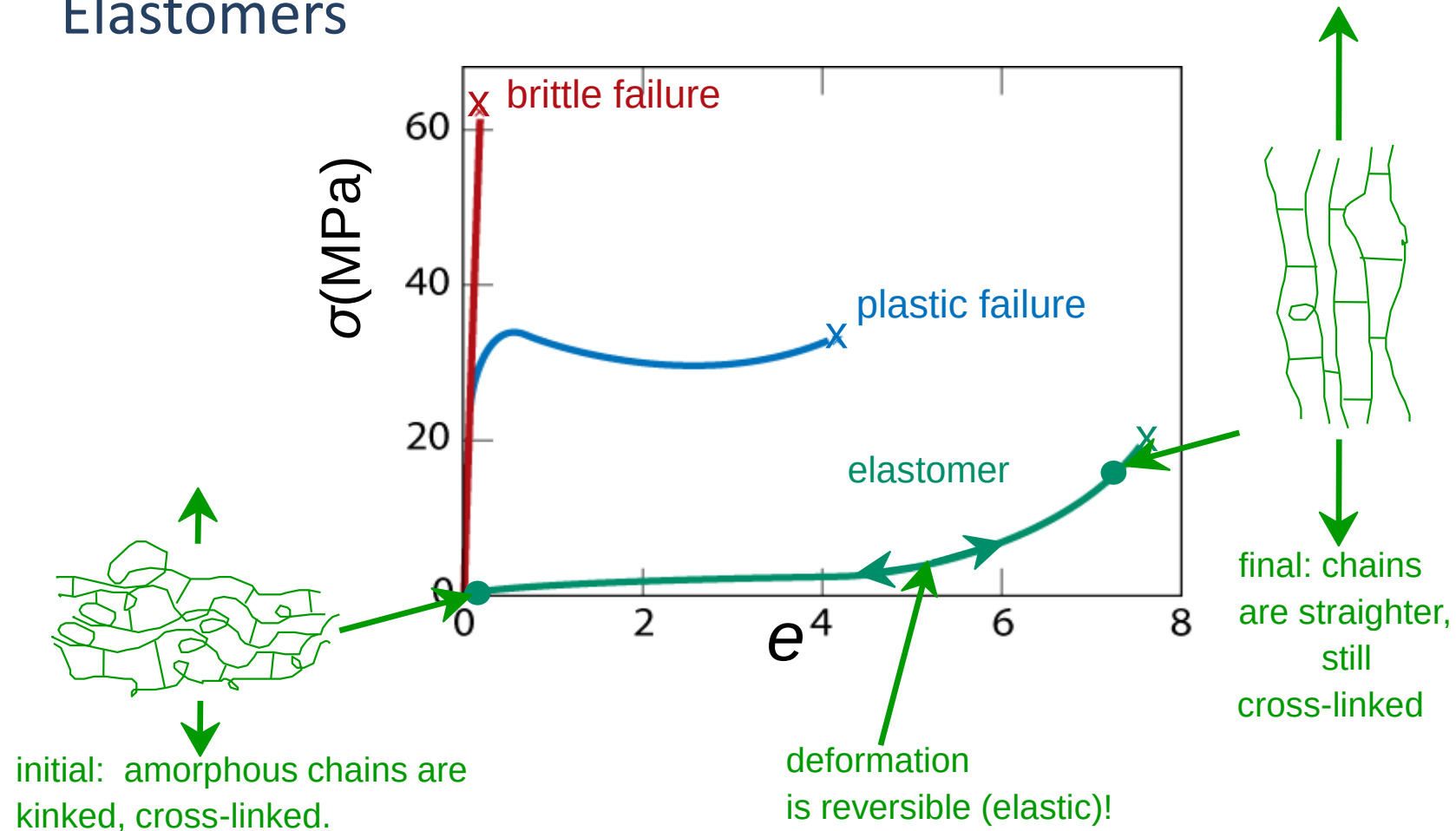


Fig 15.1 Callister & Rethwisch 9e

Mechanisms of deformation

Semi-crystalline (Plastic) Polymers

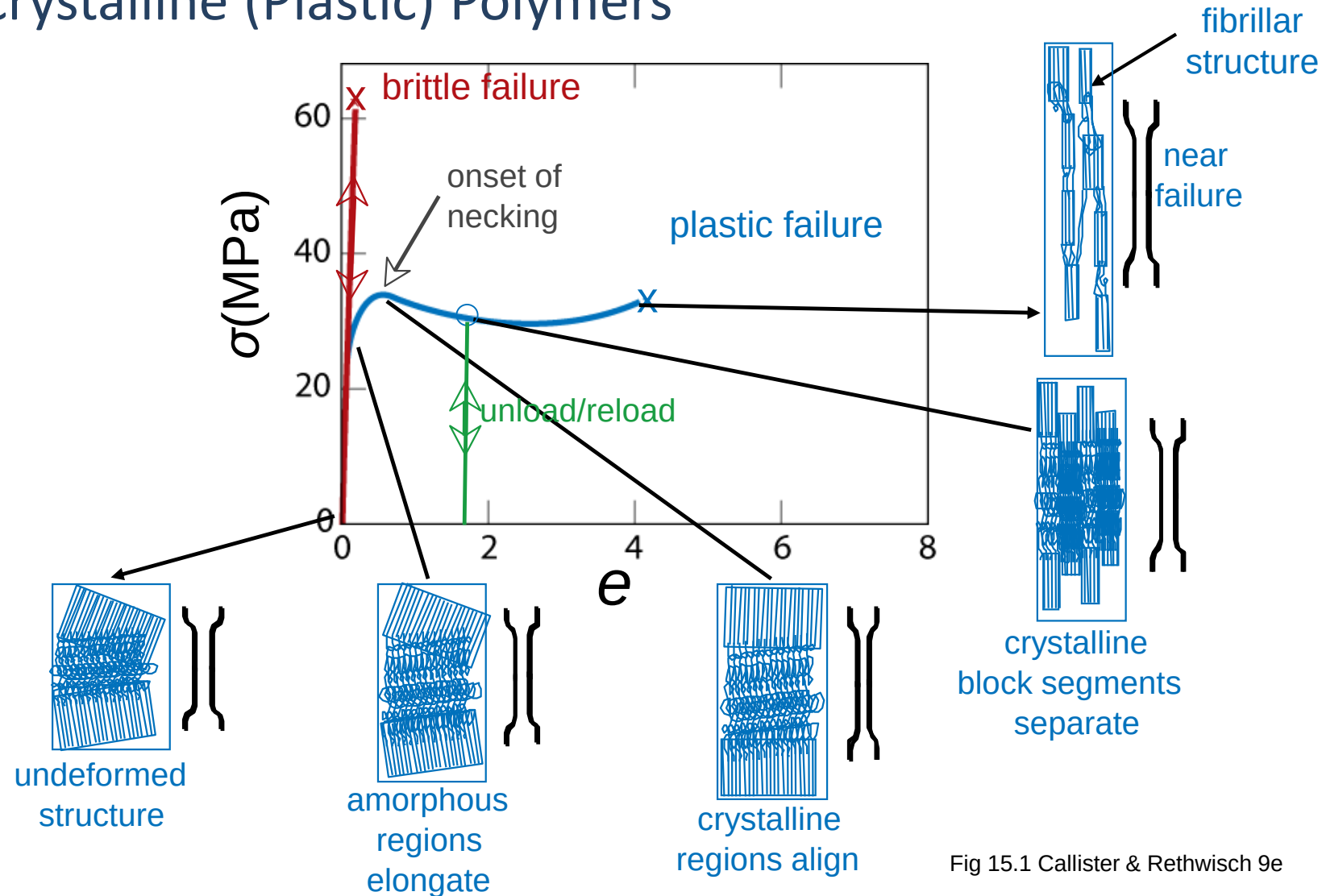
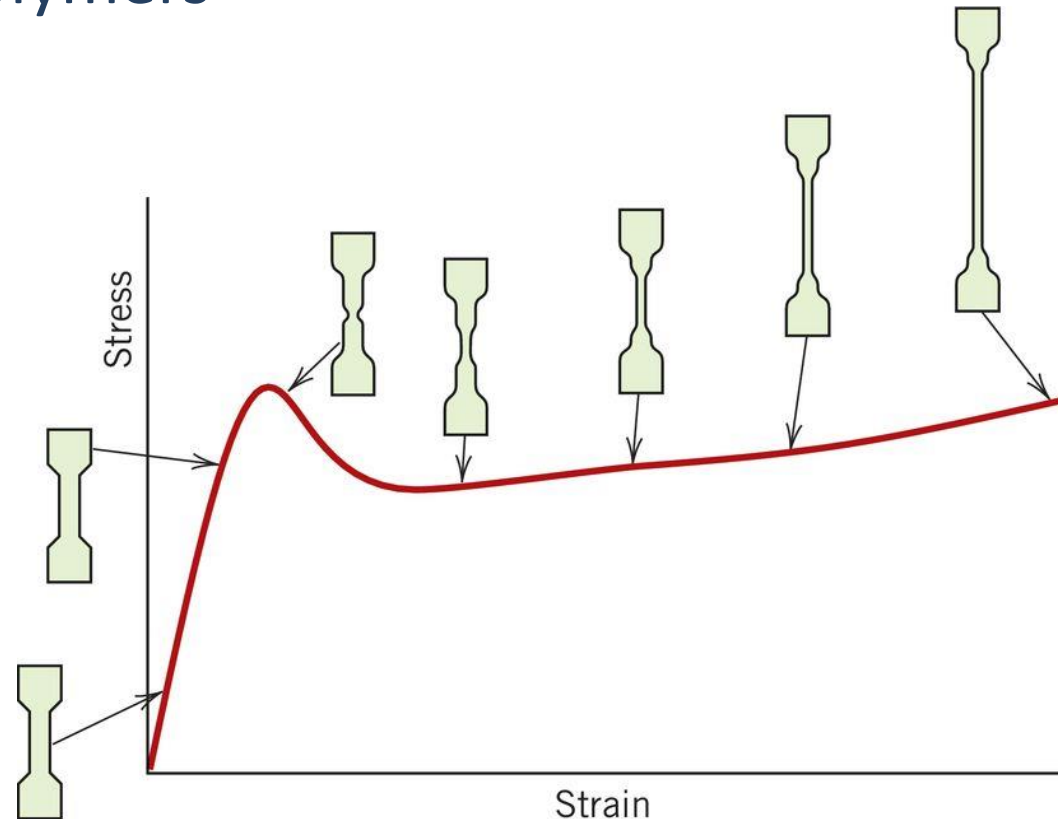


Fig 15.1 Callister & Rethwisch 9e

Stress-strain behaviour

Semi-crystalline (Plastic) Polymers

- During deformation a neck forms
- Polymer chains and crystallite regions are aligned to stress
- The neck extends along the specimen
- Fails at when it reaches the shoulder

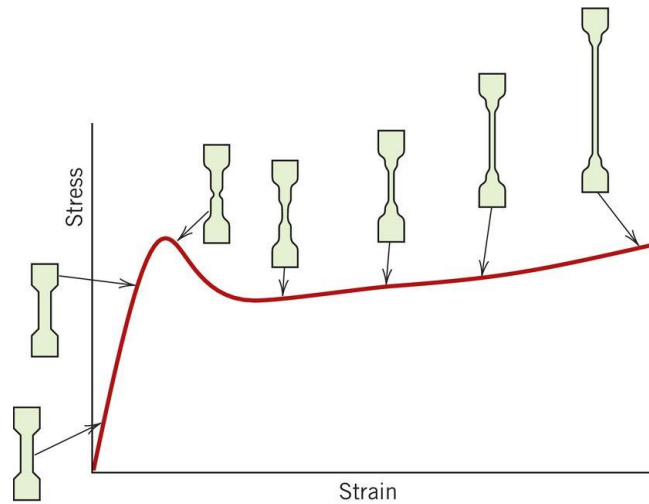


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Strain crystallization

Drawing the polymer:

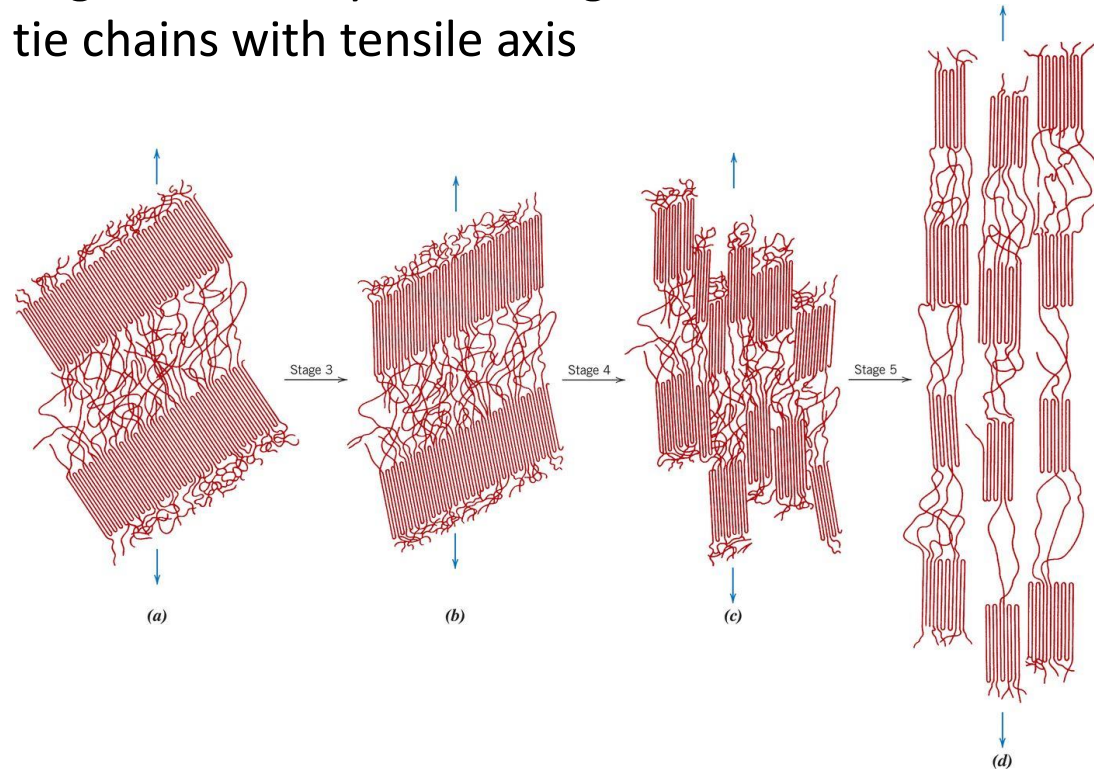
- Increases the elastic modulus (E) in the stretching direction
- Increases the tensile strength (TS) in the stretching direction
- Decreases ductility ($\%EL$)



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Fig 15.4 Callister & Rethwisch 9e

- Amorphous regions extend first
- Rotation and shear of crystalline regions
- Separation of crystalline regions
- Alignment of crystalline regions and tie chains with tensile axis



From SCHULTZ, POLYMER MATERIALS SCIENCE, 1st Edition, © 1974. Reprinted by permission of Pearson Education, Inc., Upper Saddle River, NJ.

Fig 15.13 Callister & Rethwisch 9e

Mechanical properties

Unlike metals and ceramics, **polymers do not** have constant values for their mechanical properties

The mechanical properties of polymers depend strongly on:

temperature, T

time, t

Over the range of temperature from -20°C to $+200^{\circ}\text{C}$ polymers can have a wide range of behaviour:-

Brittle elastic – plastic – viscoelastic – rubbery – viscous fluid

The behaviour depends on the ability of the large molecules to **move around and past each other** (reptate) in response to external forces

Viscoelastic deformation

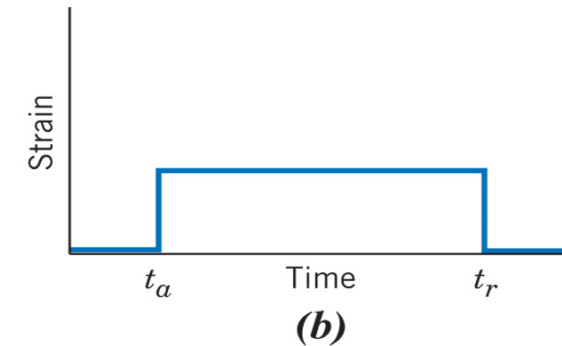
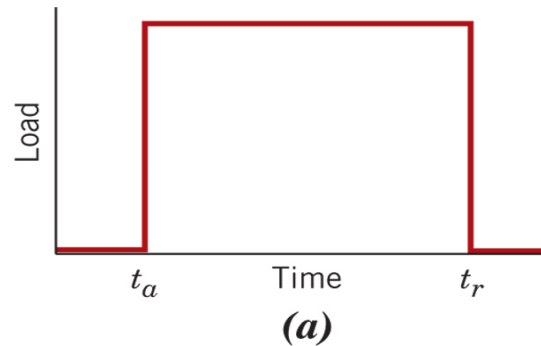
Viscoelasticity: property of materials that exhibit both viscous and elastic characteristics when undergoing deformation.

(a) Stress-strain behaviour

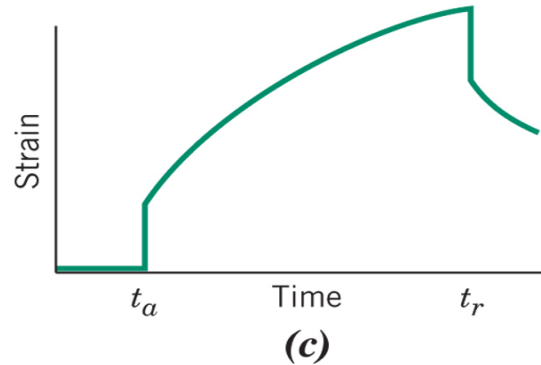
(b) Totally elastic

(c) Viscoelastic

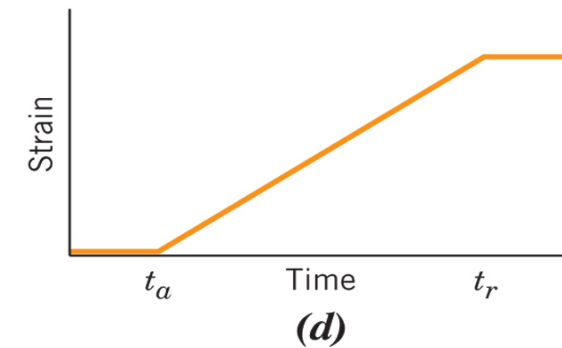
(d) Viscous



totally elastic material



viscoelastic



viscous materials

Temperature dependence

Mechanical properties should be considered relative to T_g

Elastic modulus is still the ratio of stress to strain but the strain now **depends on t and T**

$$E = \frac{\sigma}{\varepsilon(t, T)}$$

Typical behaviour for linear-amorphous polymers like polystyrene (PS)

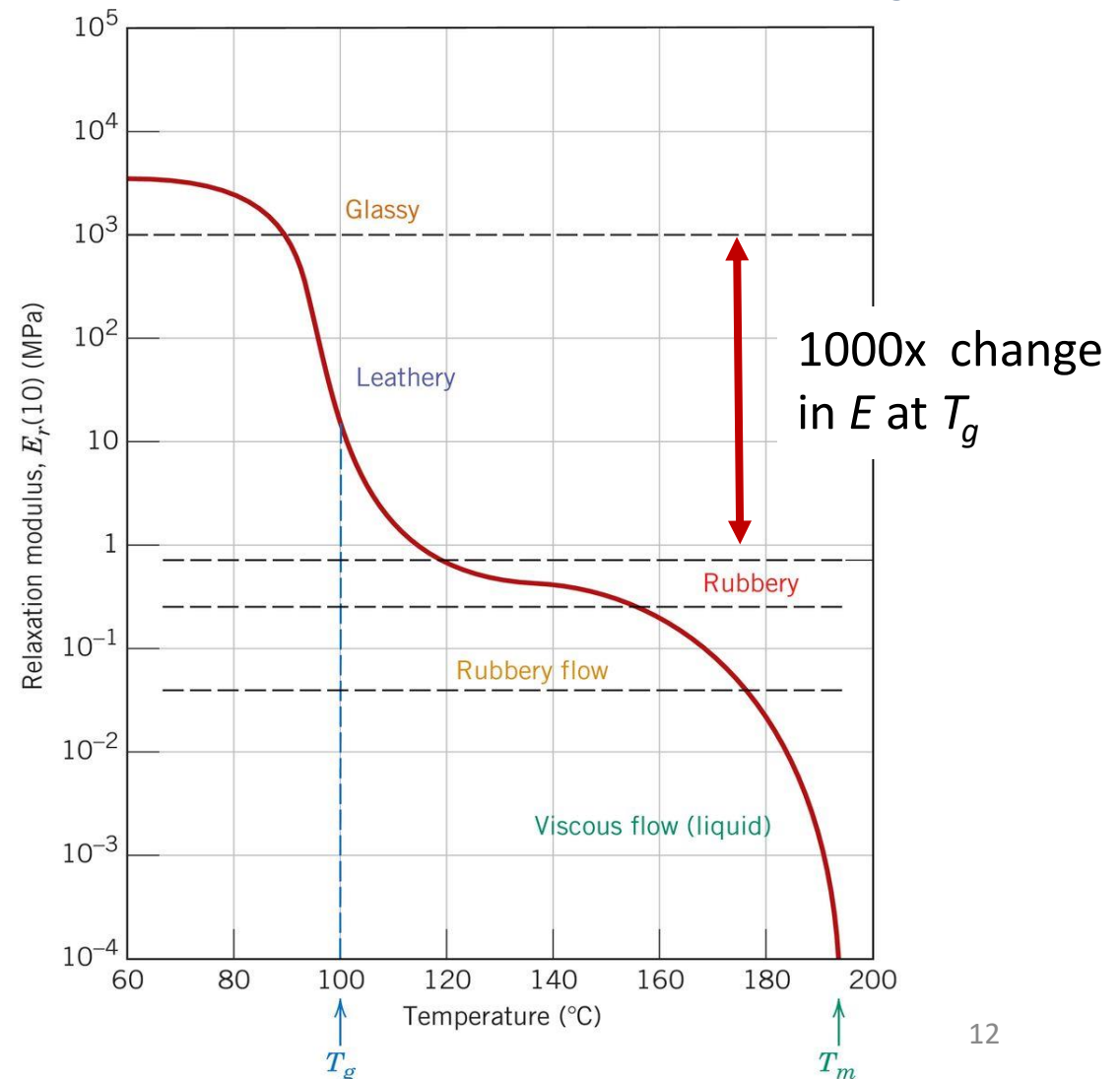
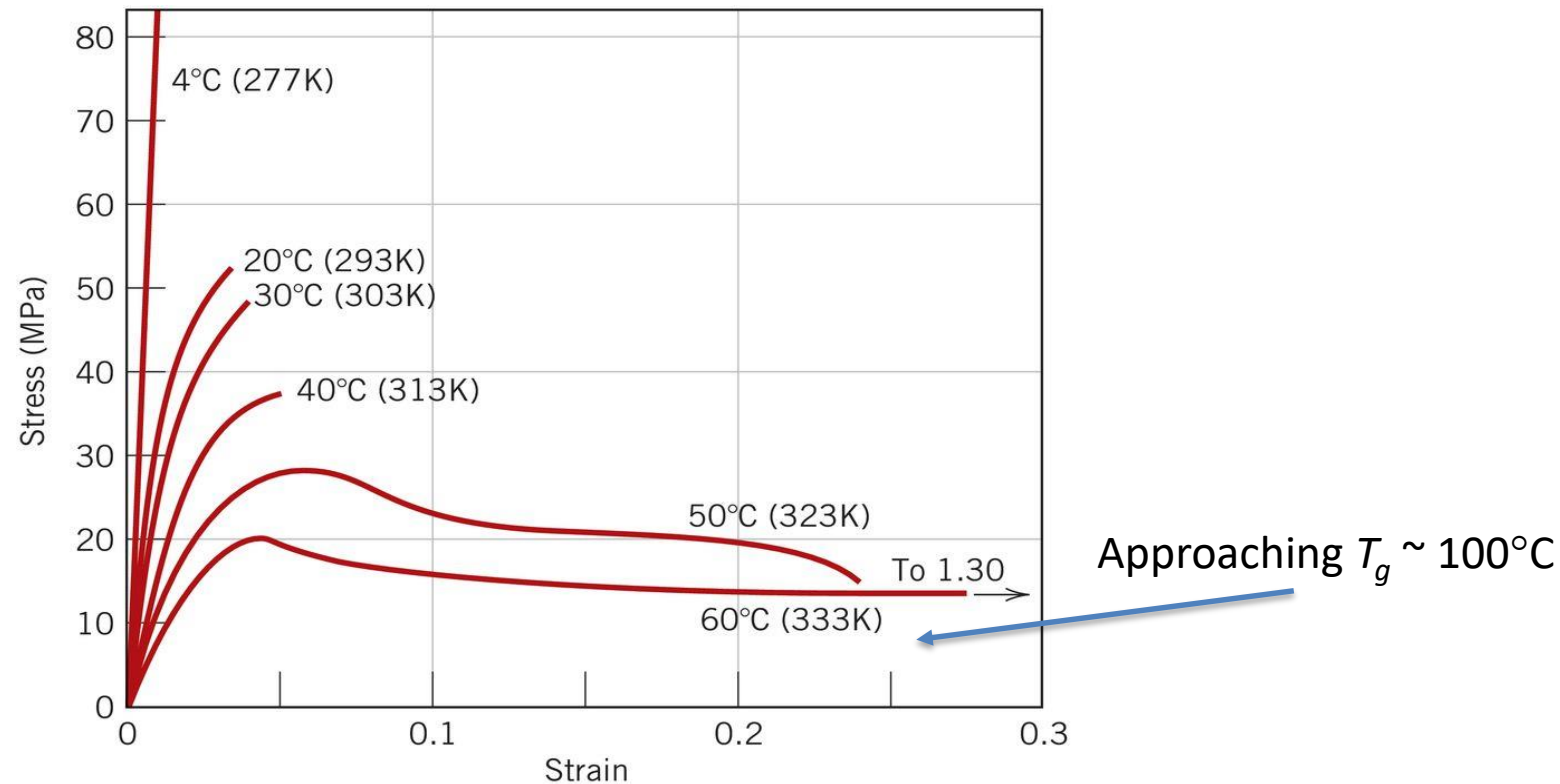


Fig 15.7 Callister & Rethwisch 9e

Temperature dependence

Stress-strain behaviour of PMMA at different temperatures



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Time dependence

Elastic modulus as a function of time and temperature

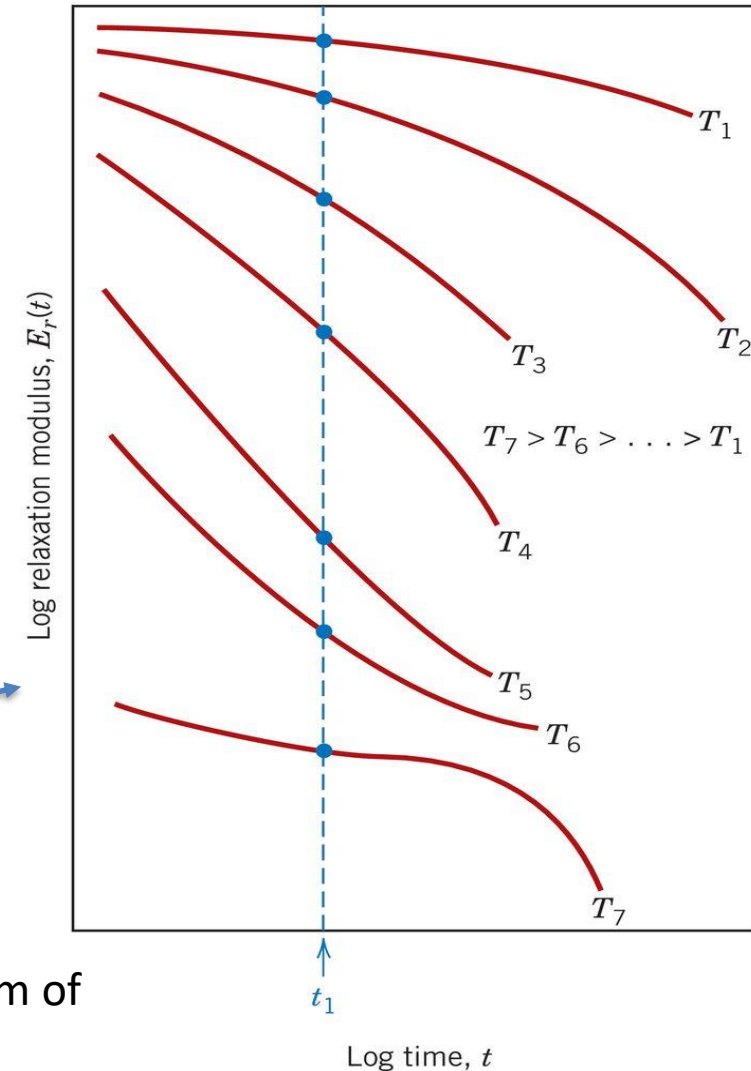
For polymers, elastic modulus is always function of time

At constant strain ε_0 the stress will decrease with time (**stress relaxation**)

The relaxation modulus is defined as

$$E_r(t) = \frac{\sigma(t)}{\varepsilon_0}$$

Relaxation modulus for different temperatures

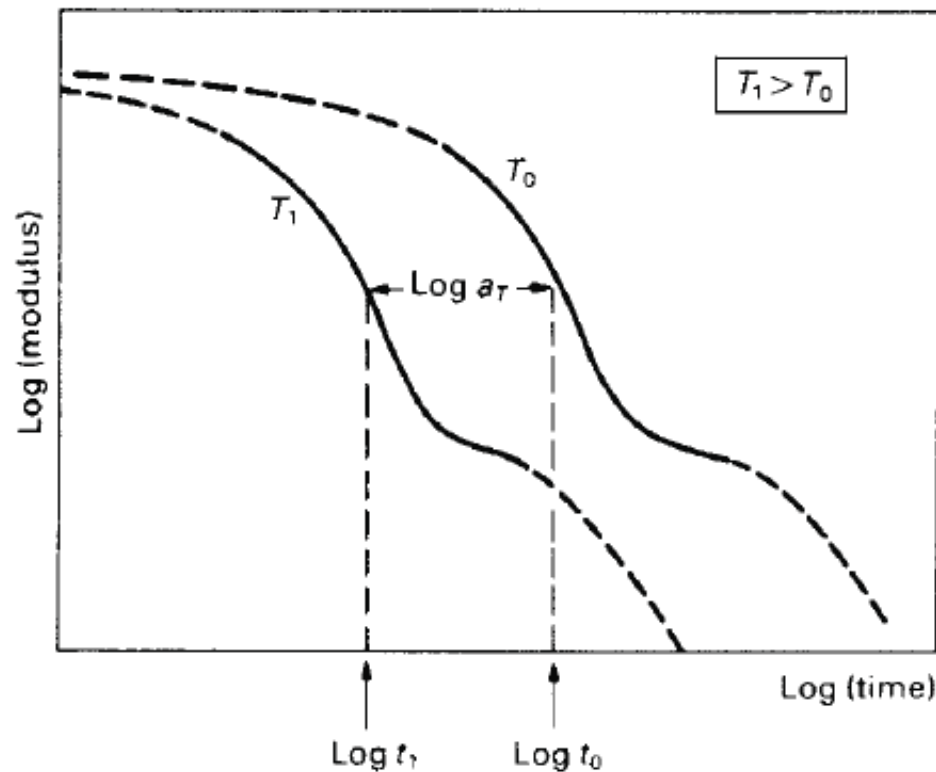


Schematic plot of logarithm of relaxation modulus versus logarithm of time for a viscoelastic polymer

Time – temperature superposition

The behaviour at different temperatures, T , is equivalent to the behaviour at different strain rates (time-scales), t

The difference can be described by a 'shift' in time a_T



$$\log(a_T) = \frac{C_1(T_1 - T_0)}{C_2 + T_1 - T_0}$$

C_1 and C_2 are constants

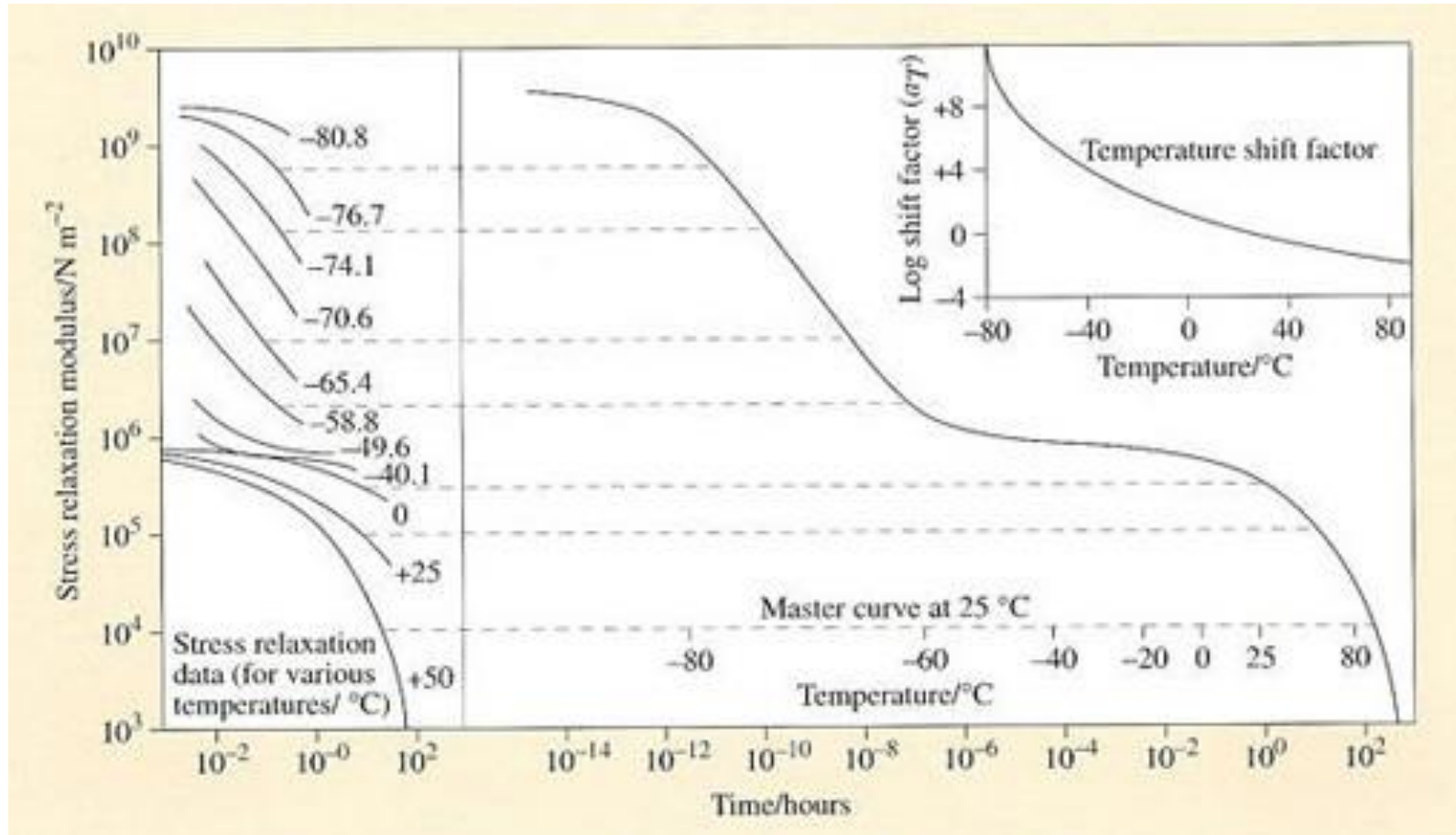
$$C_1 = 17.5 \text{ K}$$

$$C_2 = 52 \text{ K}$$

Time – temperature superposition

Master curve for time-temperature superposition

x-axis is $\log a_T / t$



Plastic flow



At $T > 1.3T_g$

Polymers at these temperatures **flow** more easily

The viscosity, η , is described in terms of shear stress, τ , and shear rate $\dot{\gamma}$

$$\eta = \frac{\tau}{\dot{\gamma}} \quad \text{units of Pa s}$$

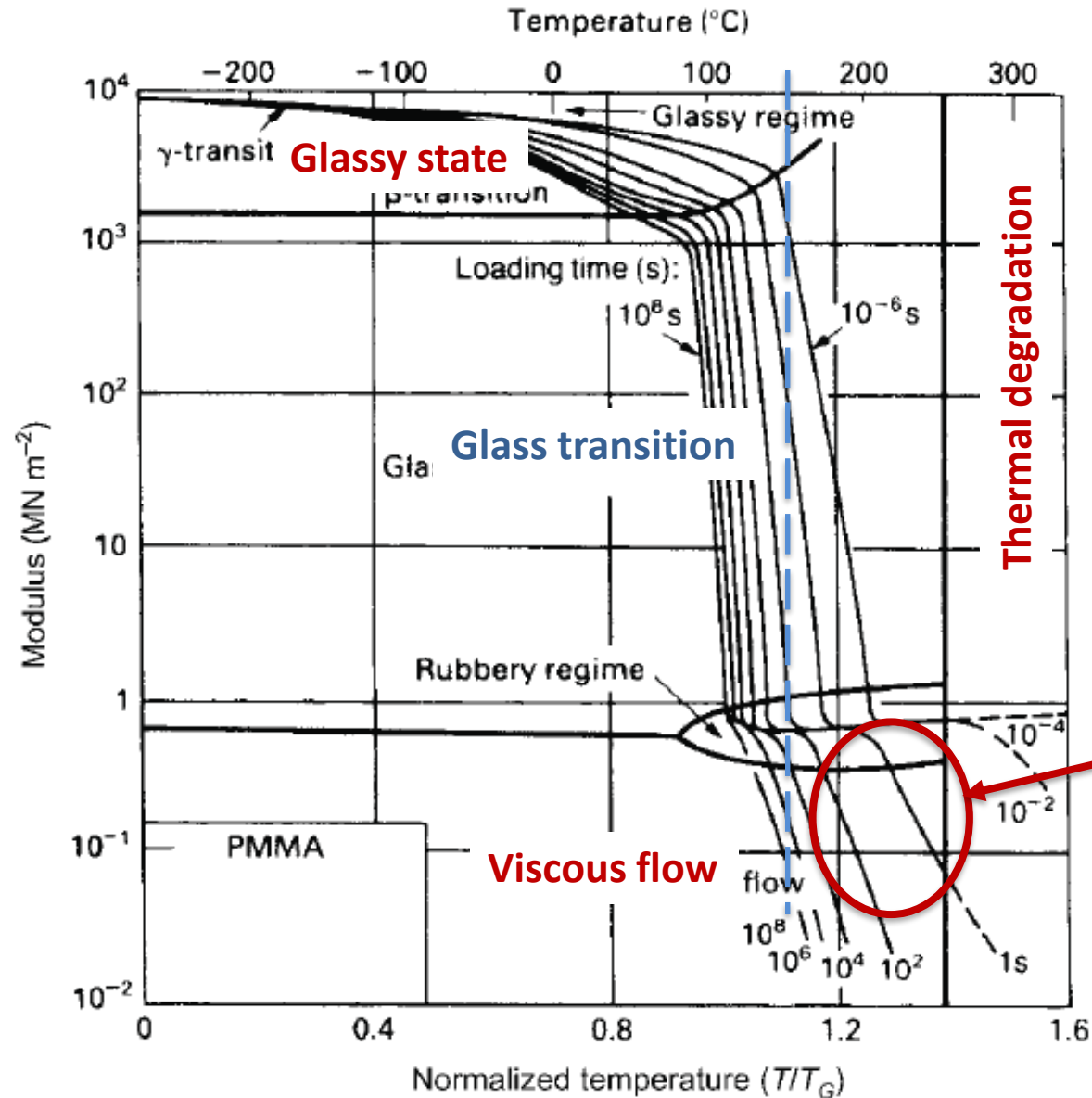
Polymers are usually moulded in the viscosity range 10^3 - 10^5 Pa s
and in the temperature range $1.3 - 1.5 T_g$

The viscous flow in this range at different temperatures can be superposed
So that if the viscosity at T_0 is η_0 , then at a higher temperature T_1

$$\eta_1 = \eta_0 \exp \left\{ -\frac{C_1(T_1 - T_0)}{C_2 + T_1 - T_0} \right\}$$

This is used to design polymer moulding processes ($F \propto \eta$)

Mechanical behaviour



Modulus diagram for PMMA

Processing temperature
and load rate **here**

Polymers are different!

Their properties depend on time and temperature

Their mechanical behaviour changes above and below T_g

We can understand this behaviour to:

- Design polymer materials and products
- Understand how they change with temperature or strain rate
- Understand how to process them (e.g. moulding)